DVA CAPSTONE PROJECT

Zomato Analytics Dashboard

Multi-Dashboard Analysis of Indian Metro Restaurant Data

Tool: Tableau | Dataset: Zomato India | Scope: 13 Cities · 826 Restaurants · 3 Dashboards

Problem Statement

The Challenge

- Zomato lists thousands of restaurants but offers no cross-city or cross-cuisine comparison.
- Inconsistencies exist between dining ratings and delivery ratings across cities.

Business Questions

- Which cities have the best dining vs delivery experiences?
- Where do rating gaps exist — and why?

Compare Ratings

Dining vs. Delivery performance by city

Cuisine Analysis

Rating gaps per cuisine type per city

Price Insights

Menu pricing patterns and tier distribution

Geo Spread

Restaurant density across India's metros

Goal: Build a comprehensive Tableau-based analytics system to help stakeholders make data-driven decisions about restaurant performance across Indian metros.

Dataset Overview

826	13	8	₹243
Total Restaurants	Metro Cities	Cuisine Types	Avg. Item Price

Dataset Fields

City	Metro city where restaurant operates
Cuisine	Type of food (Biryani, Pizza, etc.)
Dining Rating	In-restaurant experience score (0–5)
Delivery Rating	Food delivery experience score (0–5)
Votes	Number of customer reviews
Avg Item Price	Average price of menu items (INR)
Place Name	Name of the restaurant

Cities Covered

- Ahmedabad
- Bangalore
- Chennai
- Goa
- Hyderabad
- Jaipur
- Kochi
- Kolkata
- Lucknow
- Mumbai
- New Delhi
- Pune
- Raipur

⚠ *Data Note: Geographic coordinate mismatches for Goa and Kochi were corrected manually during data preparation.*

Tools & Technologies

PRIMARY TOOL

Tableau Desktop

- Built all worksheets and 3 dashboards
- Created calculated fields (Rating Gap, Above/Below Avg)
- Geographic map with bubble sizing by restaurant count
- Applied cross-dashboard city and cuisine filters
- Designed consistent dark-themed UI across all dashboards
- Packaged as .twbx (Tableau Workbook + data source)

Calculated Fields

Rating Gap = Dining Rating – Delivery Rating
Above_Below_Avg = city classification vs overall mean

Dashboard Design

Dark theme (#0D0D0D bg), Zomato red accents, consistent layout across 3 dashboards

Mapbox Integration

Used Tableau's built-in Mapbox maps for geographic restaurant distribution

Data Source

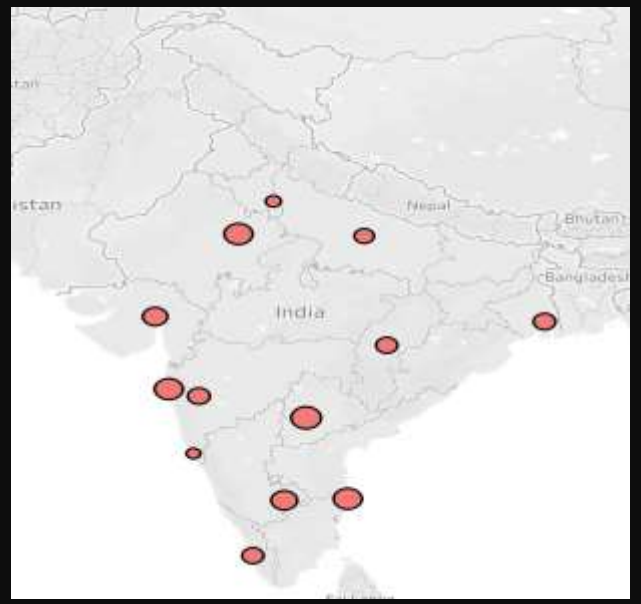
tableau_ready_zomato.csv — one flat file with all fields
Loaded as Tableau Data Extract (.tde)

01 Zomato Metro Analytics Dashboard

3.827	826	13	₹243.4
Avg Delivery Rating	Total Restaurants	Cities Covered	Average Price

Bar Chart: Avg Dining Rating by City

Map: Restaurant Distribution Across India



Insight: Raipur, Pune, and New Delhi lead in average dining ratings. Mumbai and Bangalore fall slightly below the overall average line.

02 Zomato Rating Analytics Dashboard

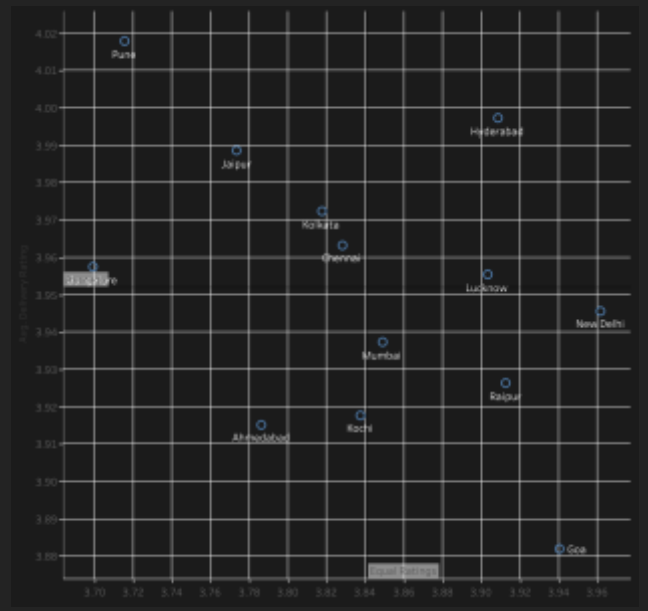
~3.80
Avg Dining Rating

3.959
Avg Delivery Rating

-0.132
Avg Rating Gap

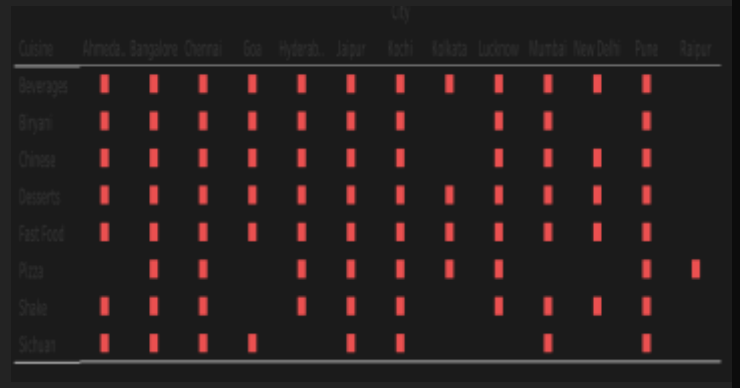
Scatter Plot

Avg Rating Comparison



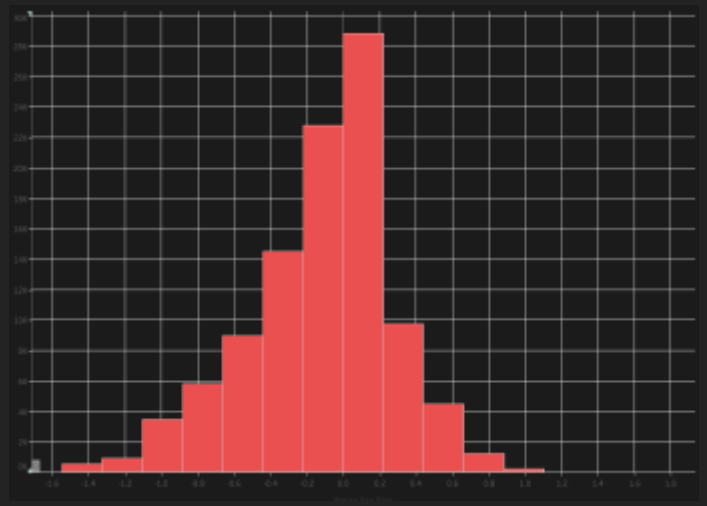
Heatmap

City-Cuisine Rating Gap



Histogram

Rating Gap Distribution



Insight: Most cities rate delivery slightly lower than dining. New Delhi scores highest on dining; Goa scores lowest on delivery. Avg Rating Gap = −0.132 (delivery trails dining).

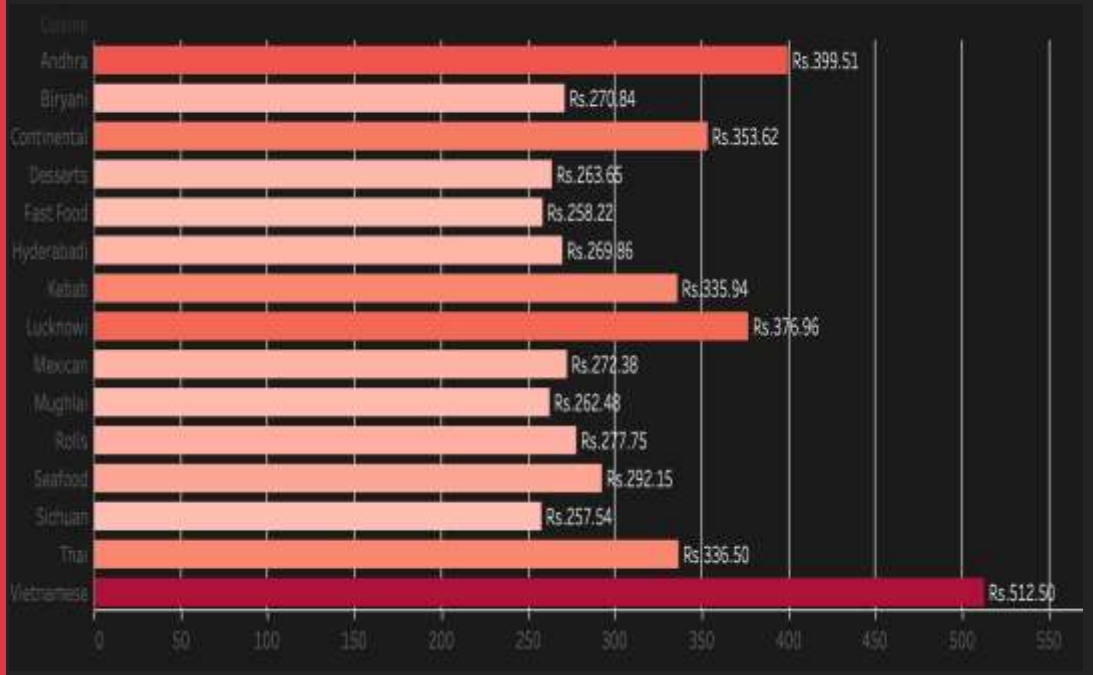
03 Zomato Menu Price Analytics Dashboard

₹243.4
Avg Item Price

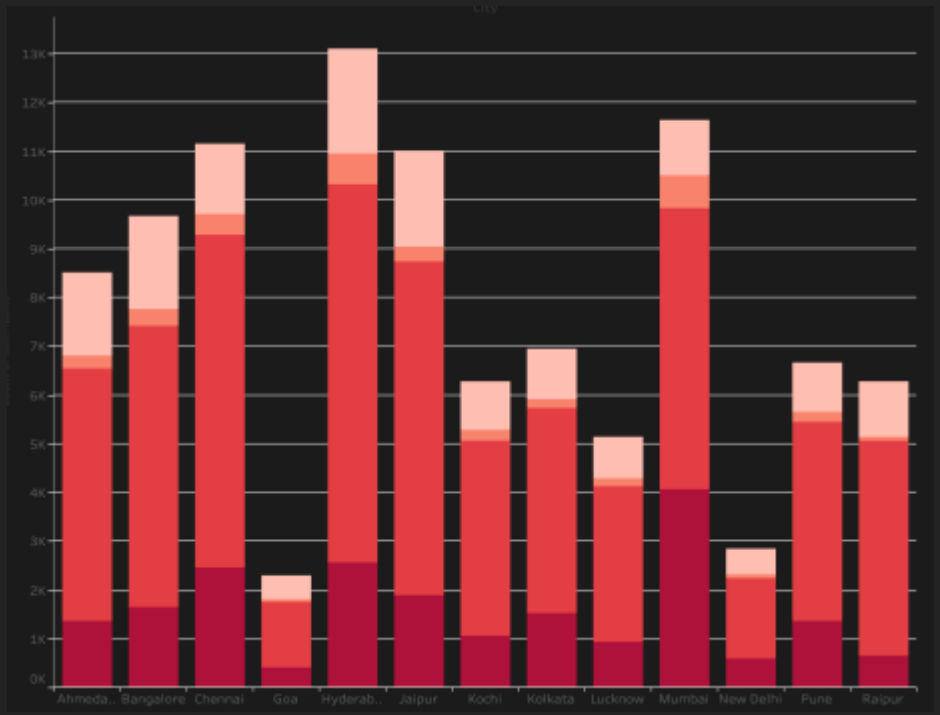
American
Highest Price Cuisine

₹3,000
Highest Item Price

Bar Chart: Avg Item Price by Cuisine



Stacked Bar: Price Tier Distribution by City



💡 Insight: Vietnamese and Andhra cuisines are the priciest. Fast Food and Desserts are most affordable. Hyderabad has the widest menu, while Goa has a smaller but pricier selection.

Chart-Level Explanation

Key Technical Choices

⚙ Calculated Field: Rating Gap

Formula: [Dining Rating] − [Delivery Rating]
Created as a custom field in Tableau.
Used in the histogram and heatmap.
Negative values = delivery exceeds dining.

Above/Below Avg Classification

Formula: IF AVG([Dining Rating]) > WINDOW_AVG(AVG([Dining Rating])) THEN 'Above Avg' ELSE 'Below Avg' END
Used for bar chart color coding.
Table calculation scope: across City.

Scatter Plot: Equal Ratings Line

A custom 'Equal Line' calculated field (value = AVG(Dining Rating)) was placed on Y-axis as a reference.
Acts as a diagonal (y=x) divider.
Cities above line: delivery > dining.

Heatmap: Square Mark Type

Chart type changed from Automatic → Square.
Color = Rating Gap magnitude.
City on columns, Cuisine on rows.
84 cells = 13 cities × 8 cuisine types.

Histogram: Bin Field

'Rating Gap (bin)' created with bin size 0.2.
Y-axis = CNT(Rating Gap).
Filter: CNT(Rating Gap) > 0 to remove null bins.
Bell-curve shape visible.

Map: Bubble Size = Restaurant Count

Mark type: Circle on map.
Size = CNT(Restaurant Name).
Detail = City.
Mapbox background (dark style).
Coordinates fixed for Goa & Kochi.

Key Insights

What the data tells us

01

Dining vs Delivery

Dining ratings are consistently higher than delivery ratings across all 13 cities. Avg Rating Gap = −0.132, meaning delivery always trails in-person experience.

02

Top Performing Cities

Raipur, Pune & New Delhi lead in avg dining ratings. Goa has the lowest avg delivery rating (3.88), suggesting delivery infrastructure challenges.

03

Rating Distribution

Most restaurants cluster near a rating gap of 0 to +0.2 (dining slightly better). Very few restaurants have extreme gaps (< −1.0 or > +1.0).

04

Cuisine Pricing

Vietnamese (₹512.50) and Andhra (₹399.51) are the most expensive cuisines. Fast Food and Desserts are the most affordable at ~₹258–264.

05

Price Distribution

Mid-tier items dominate most cities. Hyderabad has the largest menu (13K+ items). Goa and Kochi have the smallest menus but relatively higher price points.

06

Geographic Spread

Restaurant density is uneven — Mumbai, Hyderabad & Bangalore have the most restaurants. Smaller metros (Raipur, Kochi) have far fewer options available.

Challenges Faced

Geographic Mapping Errors

Problem

Goa and Kochi were misplotted on the map — appearing in wrong locations (ocean or wrong state).

Solution

Manually corrected the latitude and longitude values for both cities in Tableau's geographic role editor.

Complex Calculated Fields

Problem

The Above/Below Avg field required a WINDOW_AVG() table calculation — which has scope and direction settings that are easy to misconfigure.

Solution

Set compute direction to 'City' and tested with a crosstab to verify the classification was working correctly.

Consistent Dashboard UI

Problem

Maintaining a consistent dark theme with Zomato red accents across 3 dashboards required repeated manual formatting.

Solution

Used a shared color scheme and formatted each chart individually. Created a design reference sheet for consistency.

Cross-Dashboard Filtering

Problem

City and Cuisine filters on one dashboard needed to not accidentally affect charts on other dashboards.

Solution

Used 'Apply to worksheets' settings on each filter — selected only the worksheets inside that specific dashboard.

Conclusion

This project demonstrated how Tableau can convert raw restaurant data into actionable, visual insights. Three purpose-built dashboards — Metro Analytics, Rating Analytics, and Menu Price Analytics — together answer core business questions about Zomato's performance across Indian metro cities.

What We Achieved

- ✓ Built 3 production-ready Tableau dashboards with KPIs, filters & interactivity
- ✓ Created calculated fields (Rating Gap, Above/Below Avg) to power custom visuals
- ✓ Identified that delivery ratings consistently lag dining ratings across all 13 cities
- ✓ Revealed cuisine price hierarchy: Vietnamese most expensive, Fast Food most affordable
- ✓ Fixed data quality issues (geographic mapping errors) to ensure accurate analysis

Future Scope

- Connect to live Zomato API for real-time dashboard updates
- Add sentiment analysis of restaurant reviews (NLP)
- Build a recommendation engine using price + rating data
- Extend analysis to Tier-2 and Tier-3 cities
- Publish to Tableau Public for broader access

Thank You | DVA Capstone Project | Zomato Analytics Dashboard